Setting Up User Accounts and File Access for an E-Commerce Warehouse

SwiftCart Logistics manages an e-commerce warehouse that processes product inventory, customer orders, payment records and dispatch operations. The company is introducing a shared Ubuntu Linux server to replace files stored on separate employee computers. You have joined as its Linux System Administrator and must configure departmental access while protecting confidential records.

Use only a dedicated Ubuntu or WSL lab. Do not change unrelated users, groups or system directories. Store the practice files on the Linux filesystem under /srv, not on a Windows-mounted drive.

Employee Details

Employee	Username	Department / Group
Arun	sc_arun	Warehouse / sc_warehouse
Priya	sc_priya	Warehouse / sc_warehouse
Karthik	sc_karthik	Sales / sc_sales
Divya	sc_divya	Sales / sc_sales
Meena	sc_meena	Finance / sc_finance
Ravi	sc_ravi	Audit / sc_audit

Required Directory Structure

```
/srv/swiftcart-lab/
├── warehouse/
│   ├── products.csv
│   └── stock_updates.txt
├── sales/
│   ├── orders.csv
│   └── daily_sales.txt
├── finance/
│   ├── payments.csv
│   └── profit_report.txt
├── audit/
│   └── audit_report.txt
└── shared/
    └── announcements.txt
```

Business Access Policy

Resource	Authorized roles	Access requirement
Warehouse	Arun and Priya	Read / modify / collaborate
Sales	Karthik and Divya	Read / modify / collaborate
Finance	Meena	Read / modify; others restricted
Audit report	Ravi	Read only
Announcements	All six	Read only; admin writes
Payments exception	Ravi (temporary)	Read only via ACL; no unrelated finance access

Tasks

1 — Prepare the Company File Server

The IT manager wants all company files maintained under a common directory called swiftcart-lab inside /srv. Create warehouse, sales, finance, audit, and shared directories. Verify their existence and display their permissions.

2 — Create Departmental Groups

Create departmental Linux groups named sc_warehouse, sc_sales, sc_finance, and sc_audit. Verify that each group exists and display its group ID.

3 — Create Employee Accounts

Create Linux accounts for all six employees. Each account must have a home directory and use /bin/bash. Assign each employee to the correct departmental group. Verify UID, primary GID, supplementary groups, home directory, and login shell. Demonstrate switching from the administrator account to Arun's account.

4 — Prepare Departmental Files

Create the specified product, stock, orders, daily sales, payments, profit, audit, and announcements files using realistic sample business data. Verify their location and contents.

5 — Configure Departmental Access

Warehouse users Arun and Priya must read and update shared warehouse records; Sales users Karthik and Divya must read and update shared sales records. Meena must maintain finance files. Ravi must read but not modify the audit report. Everyone must read company announcements, but only the administrator may change them. Set suitable ownership, permissions, and collaboration behavior; prevent cross-department access.

6 — Test Unauthorized Access

As Arun, attempt to read finance/payments.csv. As Divya, attempt to update warehouse/products.csv. As Meena, read finance/payments.csv. As Ravi, attempt to modify audit/audit_report.txt. As Priya, update a warehouse file created by Arun. Record and explain each outcome.

7 — Provide Special Permission to an Auditor

Ravi needs temporary read-only access to finance/payments.csv without joining the Finance group. Ensure he cannot edit the payments file, list the finance directory, or access finance/profit_report.txt. Verify Meena retains normal access.

8 — Revoke Temporary Access

Following the audit, remove Ravi's special access to the payment file and its directory path. Prove Ravi can no longer read the file and Meena can still read and update finance records.

9 — Verify User Permissions

Show Arun's UID and groups, Ravi's group memberships, owner/group of products.csv, symbolic and numeric modes of payments.csv, ACL entries on that file, and the Warehouse and Finance directory modes. Demonstrate at least two successful operations and two denied operations.

10 — Submit the Final Access Report

Submit the directory structure, user-to-group mapping, ownership and permission report, special ACL approval and revocation evidence, test outcomes, and a brief account of problems encountered and how they were resolved.

Submission Instructions

Submit commands, relevant outputs, and terminal screenshots. Demonstrate access as the named ordinary user rather than performing all tests as root. Include permission-denied messages when expected. Explain each permission decision and document any troubleshooting.